

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Substitution: $a = \frac{(25-0)}{20}$ (1) $= 1.25 \text{ [m/s}^2\text{]} (1)$ Substitution into $F = ma$ i.e. $5\,000 \text{ [N]} = m \times 1.25 \text{ [m/s}^2\text{]} \text{ (ecf) } (1)$ $= 4\,000 \text{ [kg]} (1)$ Alternative Substitution for v and t values from the graph: $F = \frac{\Delta p}{t}$ so $F = \frac{m(25-0)}{20} (2)$ [inclusion of m (1), substitution of values (1)] $5000 \text{ [N]} = m \times 1.25 \text{ [m/s}^2\text{]} \text{ ecf } (1)$ $m = 4\,000 \text{ [kg]} (1)$ NB. Ecf on acceleration only if it is calculated [even mysteriously losing a '-' sign] Penalise mistakes in sign by -1.	1 1	1 1		4	4	
	(b)			Acceleration would be less [as mass increased] (1) [Not: slower] So the gradient would be less steep [Not: the line AB would be longer] (1)	1	1		2	1	
	(c)	(i)		If the [external] forces on an object are balanced/zero <u>resultant</u> force [accept: equal and opposite] (1) the object moves at constant velocity [or speed in a straight line] or is stationary / acceleration zero / uniform motion in a straight line (1)	2			2		
		(ii)		The velocity [accept: speed] is constant / is 25 m/s [accept: it has reached its terminal velocity / it is travelling at terminal velocity] (1) <u>Resultant</u> force [on bus] = zero / forces [on bus] are balanced (1)	1	1		2	1	
	(d)			Times at 12.5 m/s : $10 \pm 2 \text{ s}$ and $170 \pm 2 \text{ s}$ (1) [or by implication] $170 - 10 = 160 \text{ [s]} (1)$ [accept 156 to 164 s]	1	1		2	1	